

TrES-1b

R-0119

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_231014 · created 2026-07-16T04:55:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460231.7143 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.02
Transit depth (Rp/Rs)^2	2.43 +/- 0.64 [%]
Orbital Inclination (inc)	89.96 +/- 1.02
Airmass coefficient 1 (a1)	1.022 +/- 0.021
Airmass coefficient 2 (a2)	-0.0079 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	2.1 %
Best Comparison Star	#2 - [484, 79]
Optimal Aperture	4.17
Optimal Annulus	8.0
Transit Duration (day)	0.1048 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.02 vs 0.1358 (deviation 0.99σ)
Duration fit vs published	0.1048 d vs 0.1042 d (deviation 0.21σ)
Mid-time O–C	-7.4 min
Depth SNR	7.6
Residual scatter	2.1 %

Light curve

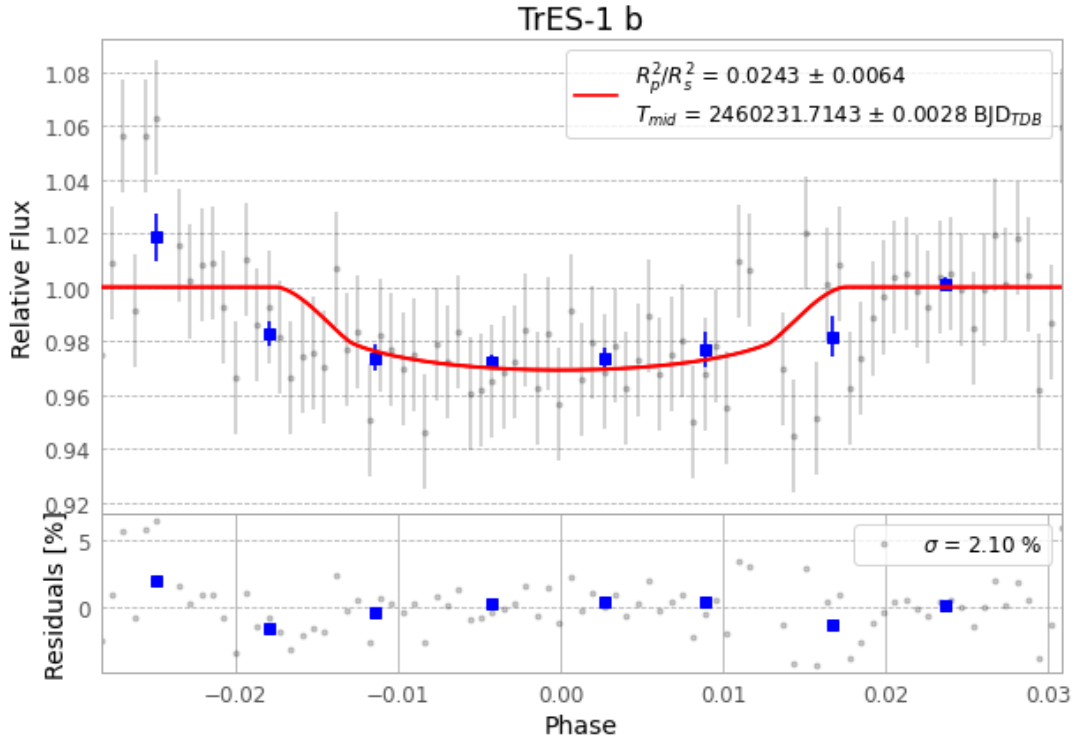


Figure 1 — FinalLightCurve_TrES-1 b_2023-10-13.png (SHA-256 9115679f746edb46...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9bc1a18b082ca8a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

91 FITS frames (60 MB), TRES-1231014030315.FITS → TRES-1231014073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-231014.log (3694c2a6eac...), FinalLightCurve_TrES-1 b_2023-10-13.png (9115679f746e...), FinalLightCurve_TrES-1 b_2023-10-13.csv (3a86ca9b900e...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 04:55Z.